

Multiple Choice (10 marks)

1. Most animal cells, regardless of species, are relatively small and about the same size. Relative to larger cells, why is this?
 - (a) Smaller cells are more resilient to external damage.
 - (b) Smaller cells have a larger number of organelles.
 - (c) Smaller cells require less energy to function.
 - (d) Smaller cells can multiply faster.
 - (e) Smaller cells fit together more tightly.
2. The average birth weight for human babies is about 3 kg. Which of the following terms best describes this?
 - (a) disruptive selection
 - (b) sexual selection
 - (c) bottleneck effect
 - (d) non-random mating
 - (e) stabilizing selection
3. Supposing that the clonal selection theory is correct, how can an organism genetically make such diverse proteins?
 - (a) Through vertical gene transfer from parent to offspring
 - (b) Through random assortment of genes during meiosis
 - (c) Through peptide bond formation errors during translation
 - (d) Through genetic mutation only
 - (e) Through transcriptional errors in RNA processing
4. Using conditioning, how can one determine the sensitivity of a dog's hearing to sounds of different frequencies?
 - (a) By using a controlled group of dogs not exposed to the sounds
 - (b) By using sensory deprivation to heighten hearing sensitivity
 - (c) By observing the dog's natural behavior in different sound environments
 - (d) By using negative reinforcement only
 - (e) By using operant conditioning
5. If a random variable X is normally distributed with a mean of 118 and a standard deviation of 11, what Z-scores correspond to raw scores of 115, 134, and 99?

- (a) -.27, 1.45, 1.73
- (b) -.27, -1.45, -1.73
- (c) -2.18, 0.15, -1.82
- (d) 0, 1.5, -2
- (e) .27, 1.45, 1.73

True / False (10 marks)

Answer True or False

- 6. True or False: Chemical shift MRI: is there any contribution to morphologic evaluation of solid breast masses?
- 7. True or False: Motor performance in chronic low back pain: is there an influence of pain-related cognitions?
- 8. True or False: Does the National Institutes of Health Stroke Scale favor left hemisphere strokes?
- 9. True or False: Should prostate specific antigen be adjusted for body mass index?
- 10. True or False: Does a colonoscopy after acute diverticulitis affect its management?

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the importance of interdisciplinary collaboration in environmental science, particularly between climatologists and geologists, in understanding the biosphere's climatic conditions.
- 12. Discuss the unidirectional nature of chemical transmission at synapses, focusing on the role of neurotransmitter distribution and synaptic vesicle presence in neurons.

End of Paper